

MOCK EXAMINATION — TRY BEFORE YOU BOOK

GCSE Chemistry

Mock Test & Mark Scheme

Time allowed	25 minutes
Total marks	35 marks
Number of questions	27
Topics covered	Atomic Structure, Bonding, Chemical Reactions, Acids & Bases

Student name: _____ **Date:** _____

Instructions

- Answer every question in the space provided on your own paper.
- Write your name and today's date at the top of your answer sheet.
- The marks for each question are shown in brackets, e.g. **(2 marks)**.
- Where a calculation is required, show your working — marks are available for method as well as the final answer.
- Once you finish, check your answers against the mark scheme at the back of this document.
- This mock test is provided free by BrainyEdHub to give you a feel for our teaching standard — book a free trial class to go through it with a tutor.

This mock test is free from BrainyEdHub — A Hub for Smarter Education. Book a free trial class at brainyedhub.com to go through your results with a tutor.

BrainyEdHub™ and the BrainyEdHub logo are trademarks of BrainyEdHub. © 2026 BrainyEdHub. All rights reserved.

Question Paper

Atomic Structure

1. State the charge of a proton. [1 mark]
2. State the charge of an electron. [1 mark]
3. Where in the atom are neutrons found? [1 mark]
4. Define 'atomic number'. [1 mark]
5. An atom has 11 protons and 12 neutrons. State its mass number. [1 mark]
6. Name the group of the periodic table that contains the noble gases. [1 mark]
7. State the number of electrons in a neutral atom with atomic number 8. [1 mark]

Bonding

8. Describe what happens to atoms when they form an ionic bond. [2 marks]
9. Name the type of bonding found in metals. [1 mark]
10. State one physical property of ionic compounds. [1 mark]
11. Describe what happens to atoms when they form a covalent bond. [2 marks]
12. Name the type of structure formed by giant covalent substances such as diamond. [1 mark]
13. State one property of substances with a simple molecular structure. [1 mark]

Chemical Reactions

14. Balance the equation: $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$. [2 marks]
15. Define 'exothermic reaction'. [2 marks]
16. State one observation you would expect to see during an exothermic reaction. [1 mark]
17. Define 'endothermic reaction'. [2 marks]
18. State the law of conservation of mass. [2 marks]
19. Name the process used to separate a soluble solid from a solution. [1 mark]
20. Name the process used to separate an insoluble solid from a liquid. [1 mark]

Acids & Bases

21. State the pH of a neutral solution. **[1 mark]**
22. State the two products formed when an acid reacts with a metal. **[2 marks]**
23. Name the salt produced when hydrochloric acid reacts with sodium hydroxide. **[1 mark]**
24. State whether a solution with pH 3 is acidic, neutral or alkaline. **[1 mark]**
25. Name the gas produced when an acid reacts with a carbonate. **[1 mark]**
26. Define 'neutralisation'. **[2 marks]**
27. Name the piece of equipment used to accurately measure the volume of acid added in a titration. **[1 mark]**

Mark Scheme

Use this guide to check your answers. Award full marks for correct final answers; where working is shown, award partial credit as indicated.

Q	Model answer / mark scheme	Marks
1	+1	1
2	-1	1
3	In the nucleus	1
4	The number of protons in an atom.	1
5	23 Mass number = protons + neutrons = 11 + 12.	1
6	Group 0 (also accept Group 8)	1
7	8 A neutral atom has equal protons and electrons.	1
8	Electrons are transferred from a metal atom to a non-metal atom, forming oppositely charged ions that attract each other.	2
9	Metallic bonding	1
10	Any valid property, e.g. high melting point, or conducts electricity when molten or dissolved.	1
11	Atoms share one or more pairs of electrons.	2
12	A giant covalent structure	1
13	E.g. low melting and boiling point, and they don't conduct electricity.	1
14	$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	2
15	A reaction that transfers energy to the surroundings, usually shown by a rise in temperature.	2
16	The temperature of the surroundings increases.	1
17	A reaction that takes in energy from the surroundings, usually shown by a fall in temperature.	2
18	Mass is neither created nor destroyed during a chemical reaction — the total mass of reactants equals the total mass of products.	2
19	Evaporation (or crystallisation)	1
20	Filtration	1
21	7	1

Q	Model answer / mark scheme	Marks
22	A salt and hydrogen gas.	2
23	Sodium chloride	1
24	Acidic	1
25	Carbon dioxide	1
26	A reaction between an acid and a base that produces a salt and water.	2
27	A burette	1

Total for Chemistry
35 marks

How did you do? A BrainyEdHub tutor can go through any question you got stuck on, and build a personalised revision plan around your results. Book a free trial class at brainyedhub.com/contact.html or message us on WhatsApp.